Sharp fractional clique packings in clique-free graphs

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Abstract

We determine the best coefficient of the excess above the Turán density in a fractional clique-packing bound under any forbidden-clique condition. For integers $r \geq 3$ and $M \geq r$, every n-vertex K_{M+1}-free graph G satisfies $\nu_r^*(G) \geq \min\{1, 2(M-1)/(r(M-r+1))\}\big(e(G) - (r-2)n^2/(2(r-1))\big)$, and the coefficient is best possible for every pair (r, M). We also determine the optimal asymptotic relative coefficient for integer K_r-packings in H-free graphs whenever $\chi(H) \geq r+1$, uniformly over all positive excesses. In particular, a fixed forbidden graph guarantees coefficient 1 precisely when its chromatic number is at most $r+2$. The proof combines fractional symmetrization with a geometric inequality for subsets of unit vectors having a positive linear dependence. A prescribed-marginal consequence of Tinarrage's sharp approximate Carathéodory theorem, together with an angle constraint arising from fractional covers, gives the required inequality for all parameters.

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1 Introduction

Turán's theorem specifies the edge threshold for forcing a clique. The corresponding packing problem asks what fraction of the excess above this threshold must be attained by edge-disjoint cliques. For arbitrary graphs the best uniform asymptotic coefficient is known. We determine exactly how that coefficient improves when the graph contains no clique of a prescribed larger order.

All graphs are finite and simple. For $r \geq 3$, let $\nu_r(G)$ be the maximum number of pairwise edge-disjoint copies of K_r in G. A fractional K_r-packing assigns nonnegative weights to copies of K_r so that the total weight on each edge is at most one; its maximum total weight is denoted by $\nu_r^*(G)$. Write $T_j(n)$ for the balanced complete j-partite graph on n vertices and $t_j(n) = e(T_j(n))$. We put $\rho_j = (j-1)/(2j)$, so $t_j(n) = \rho_j n^2 - O_j(1)$. An H-free graph contains no subgraph isomorphic to H, whether induced or not.

The packing problem has its origins in questions about decompositions. Erdős, Goodman, and Pósa [5] proved that every n-vertex graph can be decomposed into at most $t_2(n)$ edges and triangles. Such a decomposition with a triangles has $e(G) - 2a$ members, giving $\nu_3(G) \geq (e(G) - t_2(n))/2$. Král', Lidický, Martins, and Pehova [12] obtained an asymptotic bound of $n^2/2$ for the minimum total order of the members of an edge-and-triangle decomposition. Blumenthal, Lidický, Pehova, Pfender, Pikhurko, and Volec [3] subsequently determined the sharp bound and its extremal graphs for sufficiently large n. Their result yields $\nu_3(G) \geq (2/3)(e(G) - t_2(n)) - O(1)$.

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For larger cliques, Győri [7, 8] conjectured that $e(G) = t_{r-1}(n) + k$ forces $\nu_r(G) \geq (2/r - o(1))k$. Balogh and Wigal [2] proved this conjecture by establishing the finite fractional inequality

$$\nu_r^*(G) \geq \frac{2}{r}(e(G) - \rho_{r-1}n^2). \tag{1}$$

Their symmetrization reduces the fractional inequality to complete multipartite graphs. The integer conclusion uses Haxell and Rödl's theorem [10] that $\nu_r^*(G) - \nu_r(G) = o(n^2)$ uniformly, together with Győri's stronger small-excess estimate $\nu_r(G) \geq k - O_r(k^2/n^2)$ when $k = o(n^2)$.

Forbidding a larger clique can increase the coefficient. Győri and Keszegh [9] proved that every K_4-free graph with $t_2(n) + k$ edges contains at least k edge-disjoint triangles, settling a conjecture recorded by Győri [8, p. 360, Conjecture 4]. This is also related to decompositions into cliques of varying orders with cost $|V(C)| - 1$ for a member C. Balogh, He, Krueger, Nguyen, and Wigal [1] proved the conjectured asymptotic cost bound $n^2/4 + o(n^2)$ for arbitrary graphs, as well as a fractional cost bound of $t_2(n)$. In a K_4-free graph, the latter gives $\nu_3^*(G) \geq e(G) - t_2(n)$. In a recent preprint, Zhang [14] proved the exact integer cost bound $t_2(n)$ for all sufficiently large graphs. For fixed-size clique packings, however, these results do not specify the optimal dependence on the order of the forbidden clique.

For integers $r \geq 3$ and $M \geq r$, define

$$\gamma_{r,M} = \min\left\{1, \frac{2(M-1)}{r(M-r+1)}\right\}.$$

The following theorem determines the best coefficient for every forbidden-clique order. The bound is finite and has no asymptotic error.

Theorem 1.1. *For all integers $r \geq 3$ and $M \geq r$, every K_{M+1}-free graph G on $n \geq 1$ vertices satisfies*

$$\nu_r^*(G) \geq \gamma_{r,M}(e(G) - \rho_{r-1}n^2). \tag{2}$$

For each pair (r, M), the coefficient $\gamma_{r,M}$ is best possible.

The coefficient equals 1 for $M = r, r+1$, decreases for $M \geq r+1$, and tends to $2/r$ as $M \to \infty$. In terms of a forbidden K_m, it is $\min\{1, 2(m-2)/(r(m-r))\}$ for $m \geq r+1$. The assertion concerns the largest constant valid at all edge densities. It does not determine the extremal packing number separately at each density.

For fixed forbidden subgraphs, the same coefficient governs integer packings asymptotically. Here the error is relative to the excess, including excesses much smaller than n^2.

Corollary 1.2. *Fix an integer $r \geq 3$ and a graph H with $q = \chi(H) \geq r+1$. Put*

$$\beta_{r,H} = \min\left\{1, \frac{2(q-2)}{r(q-r)}\right\}.$$

For every $\varepsilon > 0$ there is n_0 such that every H-free graph G on $n \geq n_0$ vertices with $k = e(G) - t_{r-1}(n) > 0$ satisfies $\nu_r(G) \geq (\beta_{r,H} - \varepsilon)k$. The coefficient $\beta_{r,H}$ is best possible.

The chromatic number also gives an exact classification of when coefficient 1 holds. The statement includes forbidden graphs of chromatic number at most r, for which positive excesses may not exist for all large orders.

Corollary 1.3. *Fix $r \geq 3$ and a graph H. The following property holds if and only if $\chi(H) \leq r+2$. For every $\varepsilon > 0$ there is n_0 such that every H-free graph G on $n \geq n_0$ vertices with $k = e(G) - t_{r-1}(n) > 0$ satisfies $\nu_r(G) \geq (1-\varepsilon)k$.*

Remark 1.4. For each fixed M, the coefficient in (2) is strictly larger than the unrestricted coefficient in (1). For $M \geq r+1$, complete M-partite graphs with equal part sizes give equality; the endpoint $M = r$ instead uses an unbalanced r-partite graph. The finite integer theorem of Győri and Keszegh [9] is stronger than our asymptotic integer consequence when $r = 3$ and $H = K_4$. The general fractional theorem does not remove the integer-rounding error or the bounded difference between $\rho_{r-1}n^2$ and $t_{r-1}(n)$.

The new issue in the fractional argument is the dependence on the number of parts after symmetrization. Fixing a fractional cover and minimizing its excess quadratic form over the part sizes produces, at any negative minimum, a positive semidefinite matrix with a positive vector in its kernel. Its normalized Gram vectors satisfy a lower bound on their pairwise inner products. We prove that these vectors contain r distinct members whose sum has the squared norm required by (2).

The geometric input is Tinarrage's sharp approximate Carathéodory theorem [13]. We derive a version with prescribed inclusion probabilities, remove repetitions using the angle condition, and handle concentrated dependence weights by selecting their largest entries first. The last step needs both the sharp dimension-dependent variance estimate and the mean-dependent estimate in [13, Lemma 4]. These steps connect the geometric input to the fractional covering constraints; the symmetrization and the integer-transfer theorems are used as existing results.

Section 2 proves the vector inequality. Section 3 applies it to fractional covers and derives the integer consequences. Section 4 discusses the remaining questions.

2 Subset sums of unit vectors

After symmetrization, the required packing bound becomes a quadratic inequality in the part sizes. A negative minimum gives a Gram matrix with a positive vector in its kernel. We first establish a subset-sum bound that rules out such a minimum.

We use Tinarrage's sharp approximate Carathéodory theorem and its version for convex combinations with bounded coordinates [13, Theorem 3 and Lemma 7]. These statements apply to degenerate configurations by continuity, keeping the indices fixed. They also apply to s points of the unit ball in any Euclidean space after reduction to dimension $s-1$. Indeed, if c is the orthogonal projection of the origin onto their affine hull, then $\|v_i - c\|^2 = \|v_i\|^2 - \|c\|^2 \leq 1$; translation by c preserves every vector combination whose coefficients sum to zero. The mean-dependent variance estimate in [13, Lemma 4], obtained from the pivotal-sampling comparison of Chauvet and Ruiz-Gazen [4], also survives this reduction. Retaining the translated radius $\delta = \sqrt{1 - \|c\|^2}$ gives the bound $k(\delta^2 - \|\mu - c\|^2) = k(1 - \|\mu\|^2)$ for a mean μ in the affine hull; the case $\delta = 0$ is deterministic.

The next lemma retains the sharp finite-point bound while prescribing the inclusion probabilities. This distributional form is needed when some of the vectors have already been selected.

Lemma 2.1. *Let $v_1, \dots, v_s$ lie in a Euclidean unit ball, where $s \geq 2$. If $0 \leq \pi_i \leq 1$ and $\sum_i \pi_i = k$ is an integer, then there is a random k-element subset $I \subseteq [s]$ such that*

$$\mathbb{P}(i \in I) = \pi_i, \qquad \mathbb{E}\left\| \sum_{i \in I} v_i - \sum_i \pi_i v_i \right\|^2 \leq \frac{k(s-k)}{s-1}.$$

Proof. Write $V = [v_1 \ \cdots \ v_s]$, $G = V^{\mathsf{T}}V$, and $\Delta = \{x \in [0,1]^s : \sum_i x_i = k\}$. This polytope is the convex hull of the indicator vectors $\mathbf{1}_I$ of the k-element subsets. For $x \in \Delta$, let $f(x)$ be the minimum of $\sum_I p_I \|V\mathbf{1}_I\|^2$ over all probability distributions (p_I) with $\sum_I p_I \mathbf{1}_I = x$. The function f is finite,

continuous, convex, and piecewise affine. The quantity $F(x) = f(x) - \|Vx\|^2$ is the least variance among distributions with inclusion probabilities x.

We prove the bound on F by induction on s, with deterministic distributions handling $k = 0$, $k = s$, and $s = 1$. When $s = 2$ and $k = 1$, selecting index i with probability x_i gives variance $x_1x_2\|v_1 - v_2\|^2 \le 4x_1x_2 \le 1$. Now let $s \ge 3$ and choose a maximizer x^* of F on Δ. If $x_i^* = 0$ or 1, the corresponding indicator is deterministic and disappears from the centered sum. The induction hypothesis therefore applies with parameters $(s-1, k)$ or $(s-1, k-1)$. The respective differences between the required bound and the induction bounds are

$$\frac{k(k-1)}{(s-1)(s-2)} \quad \text{and} \quad \frac{(s-k)(s-k-1)}{(s-1)(s-2)},$$

both nonnegative.

Suppose that x^* is in the relative interior of Δ, and take a subgradient p of f on the affine hyperplane $\sum_i x_i = k$. For every h with $\sum_i h_i = 0$ and all sufficiently small positive and negative t, the supporting inequality gives

$$F(x^* + th) \ge F(x^*) + t(p - 2Gx^*) \cdot h - t^2 h^{\mathsf{T}} G h.$$

Maximality forces $(p - 2Gx^*) \cdot h = 0$. Hence, at each vertex $y = \mathbf{1}_I$ of Δ,

$$\|Vy\|^2 = f(y) \ge f(x^*) + 2(Gx^*) \cdot (y - x^*), \qquad \|V(y - x^*)\|^2 \ge F(x^*).$$

An optimal distribution defining $f(x^*)$ has average squared error $F(x^*)$, so the least vertex error equals $F(x^*)$. Apply [13, Lemma 7] with $d = s - 1$ and convex coordinates x^*/k. Each coordinate is at most $1/k$, and the lemma gives a vertex with squared error at most $k(s-k)/(s-1)$. This bounds the maximum of F and proves the assertion. □

Tinarrage's approximate Carathéodory theorem allows repeated indices. The angle condition arising from the covering problem lets us remove repetitions without increasing the norm.

Lemma 2.2. *Let $N \ge r \ge 3$, and let $v_1, \dots, v_N$ be unit vectors with $\langle v_i, v_j \rangle \ge -1/(r-2)$ for $i \ne j$. Suppose that $\sum_i a_i v_i = 0$ for positive weights a_i. For any r-term sum with repetitions, there is a sum of r vectors with distinct indices and no larger norm.*

Proof. Let S be such a sum, and suppose that index i occurs $m \ge 2$ times. Then $\langle S, v_i \rangle \ge m - (r-m)/(r-2) \ge 1$, so $U = S - v_i$ satisfies $\|U\|^2 \le \|S\|^2 - 1$. If an index j occurs $n_j \ge 1$ times in U, then $\langle U, v_j \rangle \ge n_j - (r-1-n_j)/(r-2) \ge 0$. The positive relation $\sum_j a_j v_j = 0$ therefore gives an unused index k with $\langle U, v_k \rangle \le 0$. There is an unused index because U has only $r-1$ terms. Replacing the deleted occurrence by v_k does not increase the squared norm and increases the number of used indices. Repetition of this step proves the lemma. □

When only two indices are omitted, an explicit distribution gives the required estimate.

Lemma 2.3. *Let $r \ge 3$, and let $v_1, \dots, v_{r+2}$ be unit vectors such that $\langle v_i, v_j \rangle \ge -1/(r-2)$ for $i \ne j$. Suppose that $\sum_i a_i v_i = 0$ for positive weights a_i with $\sum_i a_i = 1$. Then some r-element set $I \subseteq [r+2]$ satisfies*

$$\left\| \sum_{i \in I} v_i \right\|^2 \le \frac{2r}{r+1}.$$

Proof. Taking the inner product with v_i gives $a_i \le 1/(r-1)$. Order the weights decreasingly, and choose the least $p \in \{0, \dots, r-1\}$ such that $a_{p+1} + a_{p+2} \le 2B/(s-1)$, where $B = \sum_{i>p} a_i$ and $s = r + 2 - p$. Such a choice exists because the inequality is automatic when $s = 3$. Write $\ell = r - p = s - 2$ and $b_i = a_i/B$ for $i > p$. Among these s indices, omit the pair $\{i, j\}$ with probability $2/(s-1) - b_i - b_j$. These probabilities are nonnegative and sum to one. If Y is the sum of the ℓ retained vectors, then $\Pr(i \text{ is retained}) = \ell b_i$ and, for $i \ne j$, $\Pr(i, j \text{ are retained}) = (\ell - 1)(b_i + b_j - 1/(\ell+1))$. Consequently, with $\mu = \sum_{i>p} b_i v_i$ and $T = \sum_{i>p} v_i$,

$$\begin{aligned} \mathbb{E}Y &= \ell\mu, \\ \mathbb{E}\|Y\|^2 &= \frac{2\ell}{\ell+1} + (\ell^2 - 1)\|\mu\|^2 - \frac{\ell-1}{\ell+1}\|T - (\ell+1)\mu\|^2. \end{aligned} \tag{3}$$

For $p = 0$, the mean μ is zero, and this proves the lemma.

Suppose that $p > 0$, and put $x_i = a_i/B$ for $i \le p$, $t = \sum_{i \le p} x_i$, $Q = \sum_{i \le p} x_i^2$, and $u = \sum_{i \le p} v_i$. The preceding stopping condition fails, so $\ell a_p + (\ell + 2)a_{p+1} > 2B$. Since $a_p \ge a_{p+1}$, every $x_i > 1/(\ell+1)$. Also $x_i \le (1+t)/(r-1)$. If $\ell = 1$, then $p = r - 1$ and $\langle u, v_i \rangle \ge 0$ for $i \le p$. Since $x_i > 1/2$, choosing one tail index with weights b_i gives

$$\mathbb{E}\|u + Y\|^2 = 1 + \|u\|^2 - 2\sum_{i \le p} x_i \langle u, v_i \rangle \le 1.$$

It remains to consider $\ell \ge 2$. The bound $\sum_{i \le p} a_i \le p/(r-1)$ gives $t \le p/(\ell - 1)$, and hence $1/(\ell+1) \le x_i \le (1+t)/(r-1) \le 1/(\ell-1)$. Since $\mu = -\sum_{i \le p} x_i v_i$, dropping the last term of (3) gives an upper bound for $\mathbb{E}\|u + Y\|^2$ whose non-diagonal coefficients are $2f(x_i, x_j)$, where $f(x, y) = 1 - \ell(x + y) + (\ell^2 - 1)xy$. This function is nonpositive on $[1/(\ell+1), 1/(\ell-1)]^2$. Indeed, it is affine in each variable separately, and its corner values are $0, 0, -2/(\ell^2-1), -2/(\ell^2-1)$. We may therefore use the lower bound on each inner product to obtain

$$\mathbb{E}\|u + Y\|^2 \le \frac{2\ell}{\ell+1} + \frac{\ell-1}{r-2}\left[p - 2\ell t + (\ell+1)((r-1)Q - t^2)\right].$$

Summing $(x_i - 1/(\ell+1))(x_i - (1+t)/(r-1)) \le 0$ over $i \le p$ gives $(r-1)Q - t^2 \le (2\ell t - p)/(\ell+1)$. The bracketed term is thus nonpositive, so the expectation is at most $2\ell/(\ell+1) \le 2r/(r+1)$. Each outcome consists of the fixed p indices and ℓ distinct tail indices, proving the assertion. □

The following vector bound supplies the exact constant in Theorem 1.1. The small case with two more vectors than the desired subset is treated in Lemma 2.3. Prescribed-marginal sampling handles the remaining intermediate range.

Lemma 2.4. *Let $r \ge 3$ and $N \ge r+1$. Suppose that $v_1, \dots, v_N$ are unit vectors, $\langle v_i, v_j \rangle \ge -1/(r-2)$ for $i \ne j$, and $\sum_i a_i v_i = 0$ for positive weights a_i. Then some r-element subset $I \subseteq [N]$ satisfies*

$$\left\| \sum_{i \in I} v_i \right\|^2 \le \frac{r(N-r)}{N-1}.$$

Proof. Normalize $\sum_i a_i = 1$, and put $q = 1/(r-2)$. Taking the inner product with v_i gives $0 \ge a_i - q(1 - a_i)$, and hence $a_i \le 1/(r-1)$. If $N = r + 1$, put $T = \sum_i v_i$ and omit index i with probability $(1 - (r-1)a_i)/2$. These probabilities sum to one, and the expected squared norm of the remaining sum is $\|T\|^2 + 1 - 2\langle T, T/2 \rangle = 1$, as required.

Suppose next that $N \ge 2r - 1$. The vectors span a space of dimension at most $N - 1$. By [13, Theorem 3], applied to the origin with $d = N - 1$ and $k = r$, there is an r-term sum with repetitions

whose squared norm is at most $r(N-r)/(N-1)$. Indeed, the parameter $s = \min\{r, \lfloor N/2 \rfloor\}$ in that theorem satisfies $s(N-s) = r(N-r)$, also when $N = 2r-1$. Lemma 2.2 removes the repetitions.

It remains to consider $N = r + d \le 2r - 2$, where $d \ge 2$. Lemma 2.3 handles $d = 2$. Assume that $d \ge 3$, so $r \ge d + 2$. Choose $\lambda > 0$ such that $\sum_i \min\{1, \lambda a_i\} = r$. Let $P = \{i : \lambda a_i \ge 1\}$, put $p = |P|$ and $l = r - p$, and let $B = \sum_{i \notin P} a_i$. Then $l \ge 1$, $\lambda B = l$, and the remaining $l + d$ indices have normalized weights $b_i = a_i/B < 1/l$. If $p = 0$, Lemma 2.1 applied with probabilities ra_i gives the result directly.

For $p > 0$, put $x_i = a_i/B$ for $i \in P$, $t = \sum_{i \in P} x_i$, $U = \sum_{i \in P} v_i$, and $H = \sum_{i \in P} x_i v_i$. Saturation gives $x_i \ge 1/l$, and the mean of the remaining vectors under the weights b_i is $-H$. For $i \in P$,

$$\langle H, v_i \rangle \le q, \qquad \langle U, v_i \rangle \ge (l-1)q. \tag{4}$$

If $l = 1$, then $x_i \ge 1$ and $\langle U, v_i \rangle \ge 0$. Selecting one remaining index with probabilities b_i gives expected squared norm $\|U\|^2 + 1 - 2\langle U, H \rangle \le 1$.

Assume that $l \ge 2$. The weight bound gives $\sum_{i \in P} a_i \le p/(r-1)$, and therefore $t \le p/(l-1)$ and $x_i \le 1/(l-1)$. Set $W = lH - U$. Its coefficients $lx_i - 1$ are nonnegative, and (4) gives $\langle W, v_i \rangle \le q$. Thus

$$\|W\|^2 \le q(lt - p) \le \frac{p}{(r-2)(l-1)}. \tag{5}$$

For $l = 2$, a variance estimate of Tinarrage [13, Lemma 4] is sufficient. Put $Z = U - H$. The coefficients $1 - x_i$ lie in $[0, x_i]$, and (4) gives $\langle Z, v_i \rangle \ge 0$. Consequently $\|Z\|^2 \le \langle Z, H \rangle$ and $\|W\|^2 = \|H - Z\|^2 \le \|H\|^2$. The cited lemma selects two remaining indices with mean $-2H$ and variance at most $2(1 - \|H\|^2)$. After adding U, the expected squared norm is at most $2 - \|H\|^2 \le 2 < rd/(r + d - 1)$.

Finally, let $l \ge 3$. Apply Lemma 2.1 to the remaining indices with inclusion probabilities lb_i, and denote their sum by Y. Its mean is $-lH$, and its variance is at most $ld/(l + d - 1)$. Together with (5), this gives

$$\mathbb{E}\|U + Y\|^2 \le \frac{ld}{l + d - 1} + \frac{p}{(r-2)(l-1)} \le \frac{rd}{r + d - 1}.$$

To verify the last inequality, cancel $p = r - l > 0$. The resulting condition is $(r + d - 1)(l + d - 1) \le d(d-1)(r-2)(l-1)$. The ratio $(l + d - 1)/(l - 1)$ decreases with l, so it suffices to take $l = 3$. The resulting difference has positive coefficient $(2d + 1)(d - 2)$ in r; at $r = d + 2$ it equals

$$2d^3 - 4d^2 - 5d - 2 = (d-3)(2d^2 + 2d + 1) + 1 > 0.$$

This proves the required estimate for $d \ge 3$ and completes all parameter ranges. □

3 Clique packings

We first recall the fractional symmetrization that reduces the graph problem to at most M parts. The vector inequality then excludes a negative minimum of the covering quadratic form. We finish by transferring the bound to integer packings with a fixed forbidden subgraph.

A fractional K_r-cover assigns nonnegative weights to edges so that their sum on every K_r is at least one. Linear programming duality identifies its minimum total weight with $\nu_r^*(G)$. Balogh and Wigal's symmetrization has the following consequence with a forbidden clique retained.

Corollary 3.1. *Let $r \geq 3$ and $s \geq 1$ be integers, let c be real, and let G be a K_{s+1}-free graph on n vertices. There is a complete multipartite graph H on the same vertex set, with at most s nonempty parts, such that*

$$\nu_r^*(H) + ce(H) \leq \nu_r^*(G) + ce(G).$$

Proof. For the objective $\nu_r^* + ce$, iterate the clone-class replacements of [2, Lemma 2.1], as in the proof of [2, Lemma 2.2], to obtain a complete multipartite graph without increasing the objective. Every replacement preserves K_{s+1}-freeness, since a clique uses at most one vertex of the copied independent class, and replacing that vertex by a source vertex gives a clique of the same size in the previous graph. Thus the resulting graph has at most s nonempty parts. □

We also use the following formula, given in [2, proof of Claim 3.2].

Lemma 3.2. *If $n_1 \geq \cdots \geq n_r > 0$, then*

$$\nu_r^*(K_{n_1,\ldots,n_r}) = n_{r-1}n_r.$$

A complete multipartite graph with fewer than r nonempty parts has fractional K_r-packing number zero.

The case of at most r parts has a finite linear bound. We record this consequence of the two known tools to handle small supports.

Proposition 3.3. *For $r \geq 3$, every K_{r+1}-free graph G on $n \geq 1$ vertices satisfies $\nu_r^*(G) \geq e(G) - t_{r-1}(n)$.*

Proof. Apply Corollary 3.1 with $c = -1$, and let H be the terminal graph. If H has at most $r-1$ parts, then $e(H) \leq t_{r-1}(n)$ and $\nu_r^*(H) = 0$. If H has r parts of sizes $n_1 \geq \cdots \geq n_r$, merge its two smallest parts. The resulting $(r-1)$-partite graph has $e(H) - n_{r-1}n_r = e(H) - \nu_r^*(H)$ edges, which is at most $t_{r-1}(n)$. In both cases $\nu_r^*(H) - e(H) \geq -t_{r-1}(n)$. The reduction transfers this inequality to G. □

We apply the vector bound to a minimum of a fixed covering quadratic form. Working on the closed simplex also takes care of parts whose weights vanish at the minimum.

Proof of Theorem 1.1. The case $M = r$ follows from Proposition 3.3. Let $M \geq r+1$ and put $C = 2(M-1)/(r(M-r+1)) \leq 1$. By Corollary 3.1, it suffices to prove the bound for complete multipartite graphs with at most M parts. Write a_i for the part sizes normalized to have sum one. A fractional cover, averaged within each pair of parts, is given by nonnegative weights w_{ij} whose sum on every r-element set is at least one. Its normalized cost is $\sum_{i<j} w_{ij}a_ia_j$, and the normalized excess is $k = (1/(r-1) - \sum_i a_i^2)/2$.

Suppose that a cover violates the required bound. Fix its weights w_{ij} and minimize $F(a) = \sum_{i<j} w_{ij}a_ia_j - Ck$ on the closed simplex. Write $F(a) = a^{\mathsf{T}}Aa$, where

$$A_{ii} = \frac{C(r-2)}{2(r-1)}, \qquad A_{ij} = \frac{w_{ij} - C/(r-1)}{2} \quad (i \neq j).$$

Let $\eta < 0$ be the minimum and restrict to the positive support of a minimizer a, of size $N \leq M$. If $N \leq r$, Proposition 3.3, by rational blow-ups and continuity, gives cover cost at least k when $k > 0$. For $k \leq 0$, nonnegativity of the cover cost suffices. Since $C \leq 1$, both cases contradict $F(a) < 0$. Thus $N \geq r+1$.

The minimum conditions give $Aa = \eta\mathbf{1}$ and $h^{\mathsf{T}}Ah \geq 0$ whenever $\sum_i h_i = 0$. Let J be the all-ones matrix. Then $B = A - \eta J$ is positive semidefinite and $Ba = 0$. Indeed, writing any vector as $z = ta + h$, with $\sum_i h_i = 0$, gives $z^{\mathsf{T}}Bz = h^{\mathsf{T}}Ah \geq 0$. Its diagonal entries equal $\beta = C(r-2)/(2(r-1)) - \eta > 0$. Represent B/β as the Gram matrix of unit vectors v_i. They satisfy $\sum_i a_i v_i = 0$, and

$$B_{ij} + \frac{\beta}{r-2} = \frac{w_{ij}}{2} - \left(1 + \frac{1}{r-2}\right)\eta \geq 0.$$

Their pairwise inner products are therefore at least $-1/(r-2)$. For every r-element set I, the cover constraint yields

$$\beta \left\| \sum_{i \in I} v_i \right\|^2 \geq 1 - \frac{Cr}{2(r-1)} - r^2\eta. \tag{6}$$

Lemma 2.4 gives a set with squared norm at most $r(N-r)/(N-1) \leq L$, where $L = r(M-r)/(M-1)$. Substituting into (6) and using $C(r + (r-2)L)/(2(r-1)) = 1$ gives $1 \leq 1 + (r^2 - L)\eta < 1$, a contradiction. Linear programming duality and the multipartite reduction prove the bound.

For sharpness when $M \geq r+1$, take the complete M-partite graph with all part sizes equal to t. Uniform weights on its copies of K_r give a fractional decomposition, so the packing number and the excess are respectively

$$\frac{M(M-1)t^2}{r(r-1)} \quad \text{and} \quad \frac{M(M-r+1)t^2}{2(r-1)}.$$

Their ratio is $\gamma_{r,M}$. When $M = r$, take an r-partite graph with two parts of size t and the other $r-2$ parts of size $2t$. Its excess is t^2, obtained by merging the two smaller parts. Every K_r uses an edge between these two parts, so the fractional packing number is at most t^2. Assigning weight $(2t)^{-(r-2)}$ to every K_r attains this value. Thus the coefficient is best possible in every case. □

To obtain an error relative to the excess, we combine the dense-graph transfer with the stronger estimate for small excesses.

Proof of Corollary 1.2. The theorem of Erdős, Frankl, and Rödl [6, Theorem 1.5] gives a function $a(n) = o(n^2)$ such that every n-vertex H-free graph G can be made K_q-free by deleting at most $a(n)$ edges. Let G' be the resulting graph. Let $b(n) = o(n^2)$ be the uniform error in the Haxell–Rödl theorem [10], and put $g(n) = \rho_{r-1}n^2 - t_{r-1}(n) = O_r(1)$. Writing $\beta = \beta_{r,H}$, Theorem 1.1 with $M = q-1$ gives

$$\nu_r(G) \geq \nu_r(G') \geq \nu_r^*(G') - b(n) \geq \beta k - \beta(a(n) + g(n)) - b(n).$$

These estimates are uniform over G. Choose $h(n) = o(n^2)$ such that $a(n)+b(n)+g(n)+1 = o(h(n))$. For $k \geq h(n)$, the preceding bound gives $\nu_r(G) \geq (\beta - o(1))k$ uniformly. For $0 < k < h(n)$, Győri's theorem [7], in the form stated in [2, Theorem 1.6], gives $\nu_r(G) \geq k - O_r(k^2/n^2) \geq (1 - O_r(h(n)/n^2))k$. Since $\beta \leq 1$, both ranges give the required bound with an error tending to zero independently of G and k.

For sharpness when $q \geq r+2$, take $G = T_{q-1}(n)$ along integers divisible by $q-1$. These graphs are H-free, and the elementary edge bound $\nu_r(G) \leq e(G)/\binom{r}{2}$ gives

$$\limsup_{n\to\infty} \frac{\nu_r(T_{q-1}(n))}{e(T_{q-1}(n)) - t_{r-1}(n)} \leq \frac{2(q-2)}{r(q-r)} = \beta.$$

When $q = r+1$, take the complete r-partite graph with $r-2$ parts of size $2t$ and two parts of size t. It is H-free, and merging the two smaller parts gives $T_{r-1}(2(r-1)t)$, so its excess is t^2. Each K_r uses an edge between the two smaller parts, of which there are only t^2. Its packing number is therefore at most t^2, showing that the coefficient cannot exceed 1. □

Proof of Corollary 1.3. If $\chi(H) \le r+2$, choose a fixed complete $(r+2)$-partite graph F containing H, with all its parts nonempty. Every H-free graph is F-free. Since $\chi(F) = r+2$, Corollary 1.2 applied to F gives the asserted coefficient 1. Conversely, if $q = \chi(H) \ge r+3$, that corollary has the best coefficient $2(q-2)/(r(q-r)) < 1$, so the coefficient-1 property fails. This proves the equivalence, including the range $\chi(H) \le r$ not covered by the coefficient corollary itself. □

4 Concluding remarks

Theorem 1.1 determines the optimal uniform linear coefficient for every pair of clique orders. The corresponding fixed-density problem contains more information. For example, under the K_{r+1}-free restriction, the optimal uniform coefficient is 1, although the optimal density bound is strictly larger than the excess over part of the interval. A separate paper [11] determines that density function and a finite integer form for K_4-free triangle packings. It also treats the complete triangle density curve in K_5-free graphs. These results are not used in the proofs here. An earlier publicly circulated version of that paper included the cases $M = r+1$ and $r = 3$ of our coefficient theorem, together with the coefficient-1 chromatic criterion. Those results are incorporated into the general treatment here; the revised companion paper concentrates on density bounds and finite refinements. For larger forbidden cliques, determining the entire density function remains open.

The normalization by $\rho_{r-1}n^2$ cannot always be replaced by $t_{r-1}(n)$ in the finite theorem. The graph K_5, which is K_6-free, has $\nu_4^*(K_5) = 5/3$, whereas $e(K_5) - t_3(5) = 2$. For triangles, the same graph has $\nu_3^*(K_5) = 10/3 < (8/9)(e(K_5) - t_2(5))$. Thus even when the uniform coefficient is known, finite corrections require additional information.

The equality constructions establish sharpness but do not classify all equality cases. A stability theorem would need to describe near-extremal fractional covers as well as the part sizes. In the integer problem, Corollary 1.2 leaves an error $o(k)$; determining the optimal smaller error for individual forbidden graphs is another question. The vector inequality also suggests studying which angle and dependence assumptions permit sharp subset bounds for other fractional packing problems.

Declaration of AI assistance. OpenAI Codex was used during the research and preparation of this manuscript to propose and develop proof arguments, check algebraic steps and parameter ranges, assist with literature searches, and draft and revise the exposition. This included work on the subset-selection arguments and their application to fractional covers. The mathematical justification is given by the analytic proofs and the cited results; no enumeration is used as a substitute for proof.

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